Absence of crossings up to doubly exponential scales in the Cox Boolean model on Poisson lines

Working note

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Abstract

We consider the Boolean model with unit disks centred at the points of a Cox process of intensity λ per unit length on an isotropic Poisson line process of intensity μ in the plane. We prove, without any unproved hypothesis, that for small λ the model has no crossings of annuli on all scales up to a doubly exponential function of $1/\lambda$: there are constants j_0, α_0 depending only on μ such that, with $\alpha_n = \alpha_0^{(3/2)^n}$, the probability of a crossing of the annulus at scale α_n is at most $1/\alpha_n$ whenever $\lambda \le 1/(68(j_0 + 10n))$. The proof is a two-scale recursion in the manner of Gouéré with three ingredients: the occupied boxes are dominated by the union of the boxes met by at least j lines and an independent Bernoulli field; two regions of size α at distance $M\alpha$ can only share lines whose direction lies in a window of width of order $1/M$, and the remaining lines are independent between the regions; and the lines of the window are so sparse that their effect is absorbed by lowering the level j by a bounded amount. The price of the last step is a drift of the level by ten per scale, which is why the result stops at doubly exponential scales and does not yet give a subcritical phase. We state exactly what would remove the drift, show that one deterministic line costs no level in the recursion, and reduce the drift-free recursion to a two-replica estimate for one added line, which remains open.

1 Result

Let $\mathcal{L}$ be a Poisson line process in $\mathbb{R}^2$ with intensity measure $\nu(\,\mathrm{d}\theta\,\mathrm{d}p) = (\mu/\pi)\,\mathrm{d}\theta\,\mathrm{d}p$, where a line is parametrised by its normal direction $\theta \in [0,\pi)$ and its signed distance $p \in \mathbb{R}$ from the origin; the lines meeting a convex set of perimeter P form a Poisson variable with mean $\mu P/\pi$. Conditionally on $\mathcal{L}$, let X be a Poisson process of intensity λ per unit length on every line of $\mathcal{L}$, and let $\Xi := \bigcup_{c\in X} B(c,1)$. A *chain* is a finite sequence of points of X with consecutive points at distance less than 2; the disks of a chain form a connected subset of Ξ, and every connected component of Ξ is a union of chains.

Theorem 1.1 (No crossings up to doubly exponential scales)**.** *For every $\mu > 0$ there are $\alpha_0 = \alpha_0(\mu) \ge 40$ and $j_0 = j_0(\mu) \in \mathbb{N}$ such that the following holds. Let $\alpha_n := \alpha_0^{(3/2)^n}$ for $n \ge 0$. If*

$$\lambda \le \frac{1}{68\,(j_0 + 10n)}\,, \tag{1.1}$$

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then for every $x \in \mathbb{R}^2$ the probability that there is a chain from a point of $B(x, \alpha_n - 1)$ to a point outside $B(x, 8\alpha_n + 1)$ is at most $1/\alpha_n$. In particular, under (1.1), the probability that the connected component of Ξ containing a given point reaches distance $8\alpha_n + 3$ from it is at most $1/\alpha_n$.

For fixed λ the theorem applies to the scales $n \le n(\lambda) := \lfloor (1/(68\lambda) - j_0)/10 \rfloor$, and $\alpha_{n(\lambda)} \ge \exp\{\exp(c/\lambda)\}$ for a constant $c = c(\mu) > 0$ and small λ. The theorem does not show that $\lambda_c(\mu) > 0$: a subcritical phase requires the bound at all scales for one fixed λ. Section 6 explains where the restriction comes from and what would remove it.

The proof occupies Sections 2 to 5. All statements are proved in full.

2 Domination by thick boxes and a Bernoulli field

For $z \in \mathbb{Z}^2$ let $Q(z) := z + [-1/2, 1/2)^2$, the unit box centred at z, and let $N(z)$ be the number of lines of $\mathcal{L}$ meeting $Q(z)$, a Poisson variable with mean $4\mu/\pi$. Let $U = (U_z)_{z \in \mathbb{Z}^2}$ be independent random variables, uniform on $[0, 1]$, independent of $\mathcal{L}$. For $j \in \mathbb{Z}$ and $p \in [0, 1]$ put

$$\mathcal{T}_j := \{z \in \mathbb{Z}^2 \colon N(z) \ge j\}, \qquad \omega_p := \{z \in \mathbb{Z}^2 \colon U_z \le p\}, \qquad \mathcal{O}_{j,p} := \mathcal{T}_j \cup \omega_p \,, \tag{2.1}$$

so that $\mathcal{T}_j = \mathbb{Z}^2$ for $j \le 0$. A *path* is a finite sequence of sites of $\mathbb{Z}^2$ with consecutive sites at sup-distance at most 2. Distances between a site and a point of $\mathbb{R}^2$ are Euclidean.

Definition 2.1 (Crossing events)**.** For $x \in \mathbb{R}^2$, $\rho \ge 4$, $j \in \mathbb{Z}$ and $p \in [0, 1]$ let $G_{j,p}(x, \rho)$ be the event that there is a path $z_1, \dots, z_r$ of sites of $\mathcal{O}_{j,p}$ with $|z_1 - x| \le \rho$, $|z_r - x| > 8\rho$ and $|z_i - x| \le 8\rho$ for $i < r$. Put $f_{j,p}(\rho) := \sup_{x \in \mathbb{R}^2} \mathbb{P}(G_{j,p}(x, \rho))$.

All sites of such a path satisfy $|z_i - x| \le 8\rho + 3$, since a step has Euclidean length at most $2\sqrt{2} < 3$. Hence $G_{j,p}(x, \rho)$ is determined by the variables U_z with $|z - x| \le 8\rho + 3$ and by the lines meeting the boxes of these sites, all of which meet $B(x, 9\rho)$. The events are decreasing in j and increasing in p, and $f_{j,p} \equiv 1$ for $j \le 0$.

Lemma 2.2 (Domination)**.** *On a suitable probability space, U can be chosen such that for all $j \ge 1$ and $p \ge \sqrt{2}\,\lambda(j-1)$, every site z with $X \cap Q(z) \ne \varnothing$ belongs to $\mathcal{O}_{j,p}$. Consequently, the probability that there is a chain from a point of $B(x, \rho - 1)$ to a point outside $B(x, 8\rho + 1)$ is at most $f_{j,p}(\rho)$.*

Proof. Conditionally on $\mathcal{L}$, the numbers of points of X in the boxes $Q(z)$ are independent Poisson variables with means λ times the total length of the chords of the lines in $Q(z)$, which is at most $\sqrt{2}\,\lambda N(z)$. Let $q_z := \mathbb{P}(X \cap Q(z) \ne \varnothing \,|\, \mathcal{L}) \le \min\{1, \sqrt{2}\lambda N(z)\}$ and let (V_z) be independent uniform variables independent of $(\mathcal{L}, X)$. Put $U_z := q_z V_z$ if $X \cap Q(z) \ne \varnothing$ and $U_z := q_z + (1 - q_z)V_z$ otherwise. Conditionally on $\mathcal{L}$ the U_z are independent and uniform on $[0, 1]$, so U is independent of $\mathcal{L}$ and has the required law, and $X \cap Q(z) \ne \varnothing$ implies $U_z \le q_z$. If moreover $z \notin \mathcal{T}_j$, then $N(z) \le j - 1$ and $U_z \le \sqrt{2}\lambda(j-1) \le p$, so $z \in \omega_p$. For the consequence, let $c_1, \dots, c_m$ be a chain with $|c_1 - x| \le \rho - 1$ and $|c_m - x| > 8\rho + 1$, and let z_i be the site with $c_i \in Q(z_i)$. Then $|z_i - c_i| \le \sqrt{2}/2 < 1$, the sites z_i are in $\mathcal{O}_{j,p}$, consecutive ones are at sup-distance less than $2 + 1$, hence at most 2, $|z_1 - x| \le \rho$ and $|z_m - x| > 8\rho$. Stopping the sequence at the first site at distance more than 8ρ from x gives a path as in Definition 2.1. □

From now on $p := 1/48$ is fixed and we write $\mathcal{O}_j$, G_j and f_j.

3 The two-scale inclusion

For $M \geq 2$ fix finite sets $K \subseteq S_M$ and $L \subseteq S_{8M}$ on the circles of radii M and $8M$ around the origin such that every point of S_M is within distance $1/4$ of K and every point of S_{8M} within distance $1/4$ of L; one can take $|K| \leq 26M$ and $|L| \leq 202M$.

Lemma 3.1 (Inclusion, after [Gou08, Lemma 3.3])**.** *Let $M \geq 2$, $\alpha \geq 4$ and $x \in \mathbb{R}^2$. Then*

$$G_j(x, M\alpha) \subseteq \bigcup_{k \in K} \bigcup_{l \in L} \left(G_j(x + \alpha k, \alpha) \cap G_j(x + \alpha l, \alpha) \right), \tag{3.1}$$

and $|(x + \alpha k) - (x + \alpha l)| \geq 7M\alpha$ for all $k \in K$, $l \in L$.

Proof. Let $z_1, \dots, z_r$ be a path as in the definition of $G_j(x, M\alpha)$. Let i_0 be the largest index with $|z_{i_0} - x| \leq M\alpha$; then $i_0 < r$, and since $|z_{i_0+1} - x| > M\alpha$ and a step has length less than 3, $|z_{i_0} - x| > M\alpha - 3$. So z_{i_0} is within distance 3 of the circle of radius $M\alpha$ around x, and there is $k \in K$ with $|z_{i_0} - x_k| \leq 3 + \alpha/4 \leq \alpha$, where $x_k \coloneqq x + \alpha k$. Since $|z_r - x_k| > 8M\alpha - M\alpha \geq 8\alpha$, there is a first index $r' \geq i_0$ with $|z_{r'} - x_k| > 8\alpha$, and $z_{i_0}, \dots, z_{r'}$ is a path realising $G_j(x_k, \alpha)$. Similarly $8M\alpha < |z_r - x| \leq 8M\alpha + 3$, so there is $l \in L$ with $|z_r - x_l| \leq \alpha$, $x_l \coloneqq x + \alpha l$; the reversed sequence $z_r, z_{r-1}, \dots, z_1$ is a path, it starts within distance α of x_l, and $|z_1 - x_l| \geq 8M\alpha - M\alpha > 8\alpha$, so stopping it at its first site at distance more than 8α from x_l realises $G_j(x_l, \alpha)$. The last assertion is the difference of the radii. □

4 Decoupling through a direction window

Lemma 4.1 (Decoupling with a level drop)**.** *Let $M \geq 4$, $\alpha \geq 6$ and $x_1, x_2 \in \mathbb{R}^2$ with $|x_1 - x_2| \geq 7M\alpha$. Then for every $j \in \mathbb{Z}$,*

$$\mathbb{P}\big(G_j(x_1, \alpha) \cap G_j(x_2, \alpha)\big) \leq f_j(\alpha)^2 + \sum_{n \geq 1} 600\, \alpha^2\, \frac{(5\mu/M)^n}{n!}\, f_{j-n}(\alpha)^2 . \tag{4.1}$$

Proof. Let A_i be the set of sites z with $|z - x_i| \leq 8\alpha + 3$, and $R_i \coloneqq B(x_i, 9\alpha)$, which contains the boxes of the sites of A_i. Let e be the direction of $x_2 - x_1$ and $D \coloneqq |x_1 - x_2|$. A line meeting both R_1 and R_2 passes within distance 9α of both centres, so its angle φ with e satisfies $D \sin\varphi \leq 18\alpha$, whence $\varphi \leq \varphi_0 \coloneqq (\pi/2) \cdot 18\alpha/D \leq 4.04/M$.

The window. Let E be the set of lines of $\mathcal{L}$ whose direction makes an angle at most φ_0 with e, and $\mathcal{L}' \coloneqq \mathcal{L} \setminus E$. No line of $\mathcal{L}'$ meets both R_1 and R_2. Therefore the three sets of lines

$$\{\ell \in \mathcal{L}' \colon \ell \cap R_1 \neq \varnothing\}, \qquad \{\ell \in \mathcal{L}' \colon \ell \cap R_2 \neq \varnothing\}, \qquad E$$

are restrictions of the Poisson process $\mathcal{L}$ to pairwise disjoint measurable sets of lines, hence independent. The families $(U_z)_{z \in A_1}$ and $(U_z)_{z \in A_2}$ are independent of each other and of the lines, since $A_1 \cap A_2 = \varnothing$.

The level drop. For a site z write $N(z) = N'(z) + N_E(z)$ with N' the number of lines of $\mathcal{L}'$ and N_E the number of lines of E meeting $Q(z)$. The lines with direction in the window that meet $Q(z)$ have measure at most $(\mu/\pi) \cdot 2\varphi_0 \cdot \sqrt{2} \leq 5\mu/M$, because the projection of a unit square on any direction has length at most $\sqrt{2}$; so $N_E(z)$ is Poisson with mean at most $5\mu/M$ and $\mathbb{P}(N_E(z) \geq n) \leq (5\mu/M)^n/n!$. Let $H \coloneqq \max\{N_E(z) \colon z \in A_1 \cup A_2\}$. Each A_i has at most $\pi(8\alpha + 4)^2 \leq 300\alpha^2$ sites for $\alpha \geq 6$, so $\mathbb{P}(H \geq n) \leq 600\alpha^2(5\mu/M)^n/n!$ for $n \geq 1$. Let $G_i'(x, \alpha)$ be

the event of Definition 2.1 at level i with N replaced by N'. On $\{H = n\}$ every site $z \in A_1 \cup A_2$ with $N(z) \geq j$ has $N'(z) \geq j - n$, so $G_j(x_i, \alpha) \cap \{H = n\} \subseteq G'_{j-n}(x_i, \alpha)$. The event $G'_{j-n}(x_i, \alpha)$ is determined by the lines of $\mathcal{L}'$ meeting R_i and by $(U_z)_{z \in A_i}$, and H by E; by the independence established above,

$$\begin{aligned}\mathbb{P}\big(G_j(x_1,\alpha) \cap G_j(x_2,\alpha)\big) &\leq \sum_{n \geq 0} \mathbb{P}\big(G'_{j-n}(x_1,\alpha) \cap G'_{j-n}(x_2,\alpha) \cap \{H = n\}\big) \\ &= \sum_{n \geq 0} \mathbb{P}(H = n)\, \mathbb{P}\big(G'_{j-n}(x_1,\alpha)\big)\, \mathbb{P}\big(G'_{j-n}(x_2,\alpha)\big)\,.\end{aligned}$$

Since $N' \leq N$, $\mathbb{P}(G'_i(x,\alpha)) \leq \mathbb{P}(G_i(x,\alpha)) \leq f_i(\alpha)$. Bounding $\mathbb{P}(H = 0) \leq 1$ and $\mathbb{P}(H = n) \leq \mathbb{P}(H \geq n)$ for $n \geq 1$ gives (4.1). □

Proposition 4.2 (Recursion)**.** *For $M \geq 4$, $\alpha \geq 6$ and $j \in \mathbb{Z}$,*

$$f_j(M\alpha) \leq 5300\, M^2 \Big[f_j(\alpha)^2 + 600\, \alpha^2 \sum_{n \geq 1} \frac{(5\mu/M)^n}{n!}\, f_{j-n}(\alpha)^2 \Big]\,. \tag{4.2}$$

Proof. Union bound in (3.1) over the at most $26M \cdot 202M \leq 5300M^2$ pairs (k, l), and Lemma 4.1 for each pair. □

The recursion contains no fixed lines and no weights. The dependence between the two regions is carried by the window lines E alone, and it is absorbed by comparing level j at the large scale with the lower levels $j - n$ at the small scale, the comparison being cheap because n window lines through one box are rare.

5 The induction

Let $g(i) \coloneqq 1 + \min\{i, 24\}/4$ for integers $i \geq 0$, so that $1 \leq g \leq 7$, and put

$$A_\mu \coloneqq 5300\,(1 + 1200\, \mathrm{e}^{5\mu})\,. \tag{5.1}$$

Lemma 5.1 (Arithmetic)**.** *For all integers $j' \geq 0$, with $d \coloneqq 10$,*

$$1 - 2g(j' + d) + \tfrac{3}{2} g(j') \leq -1\,, \tag{5.2}$$

$$3 - \tfrac{m}{2} - 2g(j' + d - m) + \tfrac{3}{2} g(j') \leq -1 \qquad (1 \leq m \leq j' + d)\,, \tag{5.3}$$

$$3 - \tfrac{j'+d+1}{2} + \tfrac{3}{2} g(j') \leq -1\,. \tag{5.4}$$

Proof. (5.4): $\frac{3}{2} g(j') \leq \frac{3}{2} + \frac{3}{8} j'$, so the left side is at most $4 - \frac{d}{2} - \frac{j'}{8} \leq -1$. (5.2): if $j' \geq 24$ the left side is $1 - \frac{1}{2} \cdot 7 = -\frac{5}{2}$; if $j' < 24$ and $j' + d \leq 24$ it is $\frac{1}{2} - \frac{j'}{8} - \frac{d}{2} \leq -\frac{9}{2}$; if $j' < 24 < j' + d$ it is $\frac{1}{2} + \frac{3}{8} j' - 12 \leq \frac{1}{2} + \frac{69}{8} - 12 < -2$. (5.3): write $i \coloneqq j' + d - m \geq 0$. If $j' \geq 24$ and $i \geq 24$ the left side is $\frac{5}{2} - 3 - \frac{m}{2} \leq -1$; if $j' \geq 24$ and $i < 24$ it is $\frac{5}{2} + 9 - \frac{m}{2} - \frac{i}{2} = \frac{23}{2} - \frac{j'+d}{2} \leq \frac{23}{2} - 17$; if $j' < 24$ and $i \leq 24$ it is $\frac{5}{2} + \frac{3}{8} j' - \frac{j'+d}{2} \leq \frac{5}{2} - 5$; if $j' < 24 < i$ it is $\frac{5}{2} + \frac{3}{8} j' - \frac{m}{2} - 12 \leq \frac{5}{2} + \frac{69}{8} - \frac{1}{2} - 12 < -1$. □

Theorem 5.2 (Induction over scales)**.** *Let $\alpha_0 \geq \max\{A_\mu, 40\}$ and $j_0 \in \mathbb{N}$ be such that $f_{j_0}(\alpha_0) \leq \alpha_0^{-7}$, and let $\alpha_n \coloneqq \alpha_0^{(3/2)^n}$. Then for all $n \geq 0$ and all $j \geq j_0 + 10n$,*

$$f_j(\alpha_n) \leq \alpha_n^{-g(j - j_0 - 10n)} \leq \alpha_n^{-1}\,. \tag{5.5}$$

Proof. For $n = 0$, $f_j(\alpha_0) \le f_{j_0}(\alpha_0) \le \alpha_0^{-7} \le \alpha_0^{-g(j-j_0)}$ for $j \ge j_0$, since f_j is decreasing in j and $g \le 7$. Assume (5.5) for n, write $a := \alpha_n$, $J := j_0 + 10n$, and apply Proposition 4.2 with $\alpha = a$ and $M = a^{1/2} \ge 4$, so that $M\alpha = \alpha_{n+1}$. Let $j = J + 10 + j'$ with $j' \ge 0$. For $0 \le m \le j' + 10$ the level $j - m$ is at least J and $f_{j-m}(a) \le a^{-g(j'+10-m)}$; for $m > j' + 10$ we use $f_{j-m} \le 1$. The right-hand side of (4.2) is therefore at most

$$5300\, a^{1-2g(j'+10)} + 5300 \cdot 600 \sum_{m=1}^{j'+10} \frac{(5\mu)^m}{m!}\, a^{3-\frac{m}{2}-2g(j'+10-m)} + 5300 \cdot 600 \sum_{m>j'+10} \frac{(5\mu)^m}{m!}\, a^{3-\frac{m}{2}}\,.$$

By Lemma 5.1, each power of a in the first two terms is at most $a^{-1} \cdot a^{-\frac{3}{2}g(j')}$, and in the third term $a^{3-m/2} \le a^{3-(j'+11)/2} \le a^{-1} \cdot a^{-\frac{3}{2}g(j')}$. Since $\sum_{m\ge1}(5\mu)^m/m! \le \mathrm{e}^{5\mu}$, the total is at most

$$a^{-\frac{3}{2}g(j')} \cdot \frac{5300\,(1 + 1200\,\mathrm{e}^{5\mu})}{a} = \alpha_{n+1}^{-g(j')} \cdot \frac{A_\mu}{a} \le \alpha_{n+1}^{-g(j-j_0-10(n+1))}\,,$$

because $a \ge \alpha_0 \ge A_\mu$. □

Lemma 5.3 (Base case)**.** *For every $\mu > 0$ and every $\alpha_0 \ge 40$ there is $j_0 = j_0(\mu, \alpha_0)$ with $f_{j_0}(\alpha_0) \le \alpha_0^{-7}$, for $p = 1/48$.*

Proof. Fix x. On $G_{j_0}(x, \alpha_0)$, either some site within distance $8\alpha_0 + 3$ of x belongs to $\mathcal{T}_{j_0}$, or there is a path as in Definition 2.1 all of whose sites belong to ω_p. Removing loops, the path may be taken self-avoiding; it starts at one of at most $\pi(\alpha_0+1)^2 \le 4\alpha_0^2$ sites, and since it covers a distance of more than $7\alpha_0$ in steps of length at most $2\sqrt{2}$, it has at least $r_0 := \lceil 2.4\alpha_0 \rceil$ sites. There are at most 24^{r-1} self-avoiding paths with r sites from a given start, so the probability of the second alternative is at most $4\alpha_0^2 \sum_{r\ge r_0} 24^{r-1}p^r \le 4\alpha_0^2\, 2^{-2.4\alpha_0}$ for $p = 1/48$, which is at most $\alpha_0^{-7}/2$ for $\alpha_0 \ge 40$. The probability of the first alternative is at most $300\alpha_0^2\, \mathbb{P}(\mathrm{Poi}(4\mu/\pi) \ge j_0)$, which is at most $\alpha_0^{-7}/2$ for j_0 large enough. □

Proof of Theorem 1.1. Let $\alpha_0 := \max\{A_\mu, 40\}$ and $j_0 := j_0(\mu, \alpha_0)$ from Lemma 5.3. Given n and λ satisfying (1.1), let $j := j_0 + 10n + 1$. Then $\sqrt{2}\lambda(j-1) \le \sqrt{2}/68 < 1/48 = p$, so by Lemma 2.2 the probability in the theorem is at most $f_j(\alpha_n)$, and $f_j(\alpha_n) \le \alpha_n^{-1}$ by Theorem 5.2. For the last assertion, if the component of a point y reaches distance $8\alpha_n + 3$ from y, it contains a chain from a point within distance 1 of y to a point at distance more than $8\alpha_n + 1$ from y, and $1 \le \alpha_n - 1$. □

6 The drift, and what would remove it

Where the restriction comes from. In (4.2) the bound at level j and scale $M\alpha$ uses the levels $j - n$ at scale α, weighted by $\alpha^2(5\mu/M)^n/n!$. With $M = \alpha^{1/2}$ these weights are $\alpha^{2-n/2}$ up to constants, and they exceed 1 for $n \le 3$: it is likely that *some* box of a region is met by a window line, because there are of the order of $\mu\alpha^{1/2}$ window lines through a region and each meets of the order of α boxes. The level drop is applied to the whole region although only the boxes on the window lines are affected. As a consequence the hypothesis (5.5) at scale α_{n+1} is available only for levels that are higher by $d = 10$ than at scale α_n, and since the Bernoulli parameter in Lemma 2.2 is $\sqrt{2}\lambda(j-1)$, a fixed λ can only afford finitely many scales. Other choices of the scale ratio change the constants d and the exponents but not the drift: the quadratic term of the recursion requires $M \le \alpha^\kappa$ with $\kappa < 1$, and then window lines through a region are numerous.

An exact identity for the dependence. Let $\mathcal{S}$ be the set of lines of $\mathcal{L}$ meeting both R_1 and R_2, let $\pi_i(\mathcal{S}) \coloneqq \mathbb{P}(G_j(x_i,\alpha)\,|\,\mathcal{S})$, and let $\rho_i(\mathcal{S}) \coloneqq \pi_i(\mathcal{S}) - \pi_i(\varnothing) \geq 0$, the conditional probability that the crossing at x_i exists but would not exist without the shared lines. The events are conditionally independent given $\mathcal{S}$, so

$$\mathbb{P}(G_j(x_1,\alpha)\cap G_j(x_2,\alpha)) - \mathbb{P}(G_j(x_1,\alpha))\mathbb{P}(G_j(x_2,\alpha)) = \operatorname{Cov}(\rho_1(\mathcal{S}),\rho_2(\mathcal{S})) \leq \mathbb{E}\big[\rho_1(\mathcal{S})\rho_2(\mathcal{S})\big]\,. \tag{6.1}$$

The drift would disappear if one could show, at the same level j, that

$$\mathbb{E}\big[\rho_1(\mathcal{S})\rho_2(\mathcal{S})\big] \leq \alpha^{-c}\, f_j(\alpha)^2 + \varepsilon(\alpha) \tag{6.2}$$

with $c > 0$ and an error $\varepsilon(\alpha)$ below the target at scale $M\alpha$ divided by M^2; the recursion would then read $f_j(M\alpha) \leq 5300M^2[(1+\alpha^{-c})f_j(\alpha)^2 + \varepsilon(\alpha)]$ with no change of level, and Lemmas 2.2 and 5.3 would give a subcritical phase. The quantity $\rho_i(\mathcal{S})$ is the probability that a crossing *needs* a shared line: its path contains a site on a shared line at which, without that line, the level j is not reached. Heuristically such a site has exactly $j-1$ other lines, which has a small constant probability, and it must be joined to both boundaries of the annulus by two arms carried by different bundles, which costs the product of two one-arm probabilities; summing over the sites of the shared lines one expects $\mathbb{E}\rho_i$ to be of order $\alpha^{-1/2}f_j(\alpha)$ and (6.2) to hold with room to spare. The level drop of Lemma 4.1 is the crude bound $\rho_i(\mathcal{S}) \leq \mathbb{P}(G'_{j-H}\,|\,\mathcal{S})$, which ignores both the rarity of sites with exactly $j-1$ lines on a shared line and the two arms.

What a proof of (6.2) needs. Two things, neither of which we have. First, a one-arm estimate: the probability that a given site is joined to distance r in $\mathcal{O}_j$ should be smaller than the crossing probability at scale r by a factor of order $1/r$, and the induction hypothesis would have to carry it. Second, a way to use that the two arms from a needed site belong to different clusters once the site is removed; this is a disjoint-occurrence statement for clusters that share lines near the site, and it is the same difficulty, in its smallest instance, as the relay problem of our earlier notes. The present note shows that this is the only obstacle: everything else in the two-scale scheme is proved.

7 One added line

This section treats the smallest instance of the missing estimate: the effect of one deterministic line on the crossing probability. Proposition 7.2 shows that, at the level of the bounds produced by the recursion, this effect is contracted by a factor of order $M^{-1/2}$ per scale and hence decays like $\alpha^{-1/2}$, with no change of level. Proposition 7.3 reduces the drift-free recursion to a second-moment version of the one-line estimate. Proposition 7.2 is a statement about bounds, not about the ratio of the true probabilities, and it does not remove the drift.

For a line ℓ let S_ℓ be the set of sites z with $\ell \cap Q(z) \neq \varnothing$, let $N^\ell \coloneqq N + \mathbf{1}_{S_\ell}$, and let $G_j^\ell(x,\rho)$ be the event of Definition 2.1 with N replaced by N^ℓ. Put $F_j(\rho) \coloneqq \sup_{x\in\mathbb{R}^2}\sup_\ell \mathbb{P}(G_j^\ell(x,\rho)) \geq f_j(\rho)$. Trivially $F_j \leq f_{j-1}$; the point is to do without this loss of one level. In this section K and L are equally spaced on their circles with arc spacing in $[0.48, 0.5]$, so that $|K| \leq 13M$ and $|L| \leq 101M$.

Lemma 7.1 (Regions met by a line)**.** *Let $M \geq 100$, $\alpha \geq 4$, $x \in \mathbb{R}^2$ and let ℓ be a line. At most $28\sqrt{M}$ points $k \in K$ and at most $80\sqrt{M}$ points $l \in L$ satisfy $\ell \cap B(x+\alpha k, 9\alpha) \neq \varnothing$, respectively $\ell \cap B(x+\alpha l, 9\alpha) \neq \varnothing$.*

Proof. After translation and scaling by $1/\alpha$ we count the points of an equally spaced set on the circle of radius $R \in \{M, 8M\}$ around the origin within distance 9 of a line. Let $t \geq 0$ be the distance

of the line from the origin and measure the angle φ of a point of the circle from the normal direction of the line. The point is within distance 9 of the line if and only if $\cos\varphi \in [(t-9)/R, (t+9)/R]$. The set of $\varphi \in [0,\pi]$ whose cosine lies in a given interval of length $\delta := 18/R \le 0.18$ has measure at most $\arccos(1-\delta) = 2\arcsin\sqrt{\delta/2} \le 2.2\sqrt{\delta/2} = 6.6/\sqrt{R}$, since the density $1/\sqrt{1-u^2}$ of the angle is largest at the ends of $[-1,1]$ and $\arcsin s \le 1.1s$ for $s \le 0.3$. By the symmetry $\varphi \mapsto -\varphi$ the points in question lie on at most two arcs of total length at most $13.2\sqrt{R}$, which contain at most $13.2\sqrt{R}/0.48 + 2 \le 28\sqrt{R}$ points of the set. For $R = 8M$ this is at most $80\sqrt{M}$. □

Proposition 7.2 (One-line stability of the recursion)**.** *Let $\alpha_0 \ge 6$, $M_n \ge 100$ and $\alpha_{n+1} = M_n\alpha_n$ for $n \ge 0$. Let $\theta_n\colon \mathbb{Z} \to (0,1]$, $n \ge 0$, satisfy $f_j(\alpha_0) \le \theta_0(j)$ and*

$$\theta_{n+1}(j) \ge 5300\, M_n^2\Big[\theta_n(j)^2 + 600\,\alpha_n^2 \sum_{m\ge 1} \frac{(5\mu/M_n)^m}{m!}\,\theta_n(j-m)^2\Big] \qquad (j\in\mathbb{Z},\ n\ge 0)\,, \tag{7.1}$$

so that $f_j(\alpha_n) \le \theta_n(j)$ for all n and j by Proposition 4.2. Let $a_0 \ge 0$ be such that $F_j(\alpha_0) \le (1+a_0)\theta_0(j)$ for all j, and define

$$a_{n+1} := \frac{a_n}{\sqrt{M_n}} + \frac{2a_n + a_n^2}{M_n}\,. \tag{7.2}$$

Then $F_j(\alpha_n) \le (1+a_n)\,\theta_n(j)$ for all $n \ge 0$ and $j \in \mathbb{Z}$. If $a_0 \le 1$, then $a_n \le 2^n \prod_{i<n} M_i^{-1/2}$; for $M_i = \alpha_i^{1/2}$ this is $a_n \le 2^n(\alpha_0/\alpha_n)^{1/2}$.

Proof. Assume the claim for n and write $\alpha = \alpha_n$, $M = M_n$, $a = a_n$, $\theta = \theta_n$. Fix x and ℓ. The proof of Lemma 3.1 uses only the path, so (3.1) holds with G_j^ℓ in place of G_j on both sides. Fix $k \in K$, $l \in L$ and write $x_1 = x + \alpha k$, $x_2 = x + \alpha l$; let $h_i := 1$ if ℓ meets $B(x_i, 9\alpha)$ and $h_i := 0$ otherwise. If $h_i = 0$, then ℓ meets no box of a site of A_i and $G_j^\ell(x_i,\alpha) = G_j(x_i,\alpha)$. We repeat the proof of Lemma 4.1 with the deterministic line ℓ present. With E, $\mathcal{L}'$, N', N_E and H as there, on $\{H = m\}$ every site $z \in A_1 \cup A_2$ with $N^\ell(z) \ge j$ has $N'(z) + \mathbf{1}_{S_\ell}(z) \ge j - m$. Since ℓ is deterministic, the independence statements are unchanged, and $N' \le N$ gives

$$\mathbb{P}\big(G_j^\ell(x_1,\alpha) \cap G_j^\ell(x_2,\alpha)\big) \le \sum_{m\ge 0} \mathbb{P}(H=m)\,\mathbb{P}\big(G_{j-m}^\ell(x_1,\alpha)\big)\,\mathbb{P}\big(G_{j-m}^\ell(x_2,\alpha)\big)\,.$$

Here $\mathbb{P}(G_i^\ell(x_i,\alpha)) \le (1+a)^{h_i}\theta(i)$ by the induction hypothesis if $h_i = 1$ and by $f_i(\alpha) \le \theta(i)$ if $h_i = 0$. Hence the left side is at most $(1+a)^{h_1+h_2}\beta$, where β denotes the bracket in (7.1). By Lemma 7.1, the number of pairs (k,l) with $h_1 + h_2 = 2$ is at most $28\sqrt{M}\cdot 80\sqrt{M} = 2240M$, and the number of pairs with $h_1 + h_2 \ge 1$ is at most $28\sqrt{M}\cdot 101M + 80\sqrt{M}\cdot 13M \le 3900M^{3/2}$. Since there are at most $1313M^2 \le 5300M^2$ pairs in total,

$$\mathbb{P}(G_j^\ell(x, M\alpha)) \le \beta\big[5300M^2 + 3900M^{3/2}a + 2240M(2a+a^2)\big] \le 5300M^2\beta\Big[1 + \frac{a}{\sqrt{M}} + \frac{2a+a^2}{M}\Big]\,,$$

which is at most $(1+a_{n+1})\theta_{n+1}(j)$ by (7.1). If $a \le 1$, then $(2a+a^2)/M \le 3a/M \le 0.3a/\sqrt{M}$, so $a_{n+1} \le 2a/\sqrt{M} \le 1$, and the bound on a_n follows by induction. For $M_i = \alpha_i^{1/2}$ we have $\prod_{i<n}\alpha_i^{-1/4} = \alpha_0^{-\frac14\sum_{i<n}(3/2)^i} = (\alpha_0/\alpha_n)^{1/2}$. □

What the proposition says and what it does not. A line meets only of the order of $\sqrt{M}$ of the M regions on a circle, and of the order of M of the M^2 pairs in both regions; this is the whole mechanism. The proposition shows that whatever bounds the recursion (4.2) propagates for f_j, it propagates for F_j at the *same* levels, up to a factor $1 + a_n$ with a_n of order $\alpha_n^{-1/2}$ up to the factor 2^n, which is a power of $\log \alpha_n$. A simulation of the lattice model ($\mu = 1.4$, $j = 4$, line through the centre, $4 \le \alpha \le 24$) gave a relative excess $(\mathbb{P}(G_j^{\ell}) - \mathbb{P}(G_j))/\mathbb{P}(G_j)$ decreasing from 0.17 to 0.06, with fitted exponent about -0.4, in agreement with this order. The limitations are these. The statement compares F_j with the bound θ_n, not with f_j itself. In Theorem 5.2 the base bound is $\theta_0(j) = \alpha_0^{-7}$ for $j \ge j_0$, which does not distinguish levels, so there the proposition gives nothing beyond $F_j \le f_{j-1}$ with j_0 increased by one. The recursion (7.2) contracts only once a_n is below $\sqrt{M_n}$. And the proposition concerns one deterministic line, whereas a region carries of the order of $\mu\alpha/M$ window lines.

Why the supremum over several lines does not close. The natural extension is a hypothesis $\mathbb{P}(G_j^{\Lambda}(x, \alpha)) \le (1 + a_\alpha m)\theta(\alpha)$ uniformly in sets Λ of m lines, each raising the count by one on its boxes. The same counting gives $a_{M\alpha}$ of order $a_\alpha/\sqrt{M}$, so the hypothesis at scale $M\alpha$ is non-trivial for m up to $\sqrt{M}/a_\alpha$, while at scale α it is non-trivial only for $m \le 1/a_\alpha$. Since all m lines may pass through one sub-region, the sub-regions met by more than $1/a_\alpha$ lines can only be bounded by 1, and there may be of the order of $\sqrt{M}\, a_\alpha m$ of them; this loses a factor $1/\theta(\alpha)$. The obstruction reflects a real effect: a dense family of nearly parallel lines lowers the level by one on a whole strip, and the excess is then not expected to be linear in m. A uniform bound over line sets is therefore the wrong object, as in our earlier notes; the window lines are Poisson, and fans are improbable.

A reduction that keeps the randomness. Let x_1, x_2, R_i be as in Lemma 4.1, let $\mathcal{S}$ be the set of lines of $\mathcal{L}$ meeting both R_1 and R_2, a Poisson process whose intensity measure $\nu_{\mathcal{S}}$ has total mass at most $(\mu/\pi) \cdot 2\varphi_0 \cdot 18\alpha \le 47\mu\alpha/M$, and let $\pi_i(\mathcal{S}) \coloneqq \mathbb{P}(G_j(x_i, \alpha) \,|\, \mathcal{S})$ as in Section 6. For a line ℓ put $D_\ell \pi_i(\mathcal{S}) \coloneqq \pi_i(\mathcal{S} \cup \{\ell\}) - \pi_i(\mathcal{S}) \ge 0$, the conditional probability given $\mathcal{S}$ that the crossing at x_i exists with the additional line ℓ but not without it.

Proposition 7.3 (Second-moment reduction)**.** *In this situation,*

$$\mathbb{P}(G_j(x_1, \alpha) \cap G_j(x_2, \alpha)) \le \mathbb{P}(G_j(x_1, \alpha))\mathbb{P}(G_j(x_2, \alpha)) + \prod_{i=1,2} \Big(\int \mathbb{E}[(D_\ell \pi_i(\mathcal{S}))^2]\, \nu_{\mathcal{S}}(\,\mathrm{d}\ell) \Big)^{1/2}. \quad (7.3)$$

Consequently, if $\mathbb{E}[(D_\ell \pi_i(\mathcal{S}))^2] \le \delta^2 f_j(\alpha)^2$ for $i = 1, 2$ and $\nu_{\mathcal{S}}$-almost every ℓ, then the left side is at most $(1 + 47\mu\alpha\delta^2/M)\, f_j(\alpha)^2$, and

$$f_j(M\alpha) \le 5300\, M^2\, (1 + 47\mu\alpha\delta^2/M)\, f_j(\alpha)^2 \quad (7.4)$$

if this holds for all pairs of Lemma 3.1.

Proof. Given $\mathcal{S}$, the events $G_j(x_1, \alpha)$ and $G_j(x_2, \alpha)$ are independent, because the lines of $\mathcal{L} \setminus \mathcal{S}$ meeting R_1 and those meeting R_2 are independent of each other and of $\mathcal{S}$, and so are the families $(U_z)_{z \in A_i}$. Hence the left side of (7.3) equals $\mathbb{E}[\pi_1(\mathcal{S})\pi_2(\mathcal{S})] = \mathbb{E}\pi_1\, \mathbb{E}\pi_2 + \mathrm{Cov}(\pi_1, \pi_2)$, and $\mathrm{Cov}(\pi_1, \pi_2) \le (\mathrm{Var}\,\pi_1\, \mathrm{Var}\,\pi_2)^{1/2}$. The Poincaré inequality for functionals of a Poisson process gives $\mathrm{Var}\,\pi_i(\mathcal{S}) \le \int \mathbb{E}[(D_\ell \pi_i(\mathcal{S}))^2]\, \nu_{\mathcal{S}}(\,\mathrm{d}\ell)$. The consequences follow from the bound on the mass of $\nu_{\mathcal{S}}$ and from Lemma 3.1. □

The recursion (7.4) has no change of level. With $M = \alpha^\kappa$ it requires $\delta^2 \le \alpha^{\kappa-1}$ up to constants, i.e. a relative one-line effect below $\alpha^{-(1-\kappa)/2}$ in $L^2(\mathcal{S})$; for $\kappa = 1/2$ this is $\alpha^{-1/4}$, weaker than the order $\alpha^{-1/2}$ of Proposition 7.2 and of the simulation. What is missing is the passage from the first moment to the second: $\mathbb{E}[D_\ell \pi_i(\mathcal{S})] = \mathbb{P}(G_j^\ell(x_i, \alpha)) - \mathbb{P}(G_j(x_i, \alpha))$ is the one-line excess, but $\mathbb{E}[(D_\ell \pi_i(\mathcal{S}))^2]$ is the probability that two realisations of the model in R_i which share the lines of $\mathcal{S}$ and are otherwise independent both have a crossing that needs ℓ. The bound $D_\ell \pi_i \le 1$ only gives $\mathbb{E}[(D_\ell \pi_i)^2] \le F_j(\alpha) - \mathbb{P}(G_j(x_i, \alpha))$, which is of first order in f_j and insufficient. A subcritical phase would follow from

$$\mathbb{E}[(D_\ell \pi_i(\mathcal{S}))^2] \le \alpha^{-c} \frac{M}{\mu\alpha} f_j(\alpha)^2 \qquad \text{for some } c > 0\,, \tag{7.5}$$

a two-replica one-line estimate. We have no proof of (7.5).

8 Summary

1. Theorem 1.1 is unconditional: for $\lambda \le 1/(68(j_0 + 10n))$ the Cox Boolean model on Poisson lines has annulus-crossing probability at most $1/\alpha_n$ at the scale $\alpha_n = \alpha_0^{(3/2)^n}$, i.e. the model is subcritical on all scales up to $\exp\{\exp(c/\lambda)\}$.
2. The proof uses the domination of the occupied boxes by $\mathcal{T}_j \cup \omega_p$ (Lemma 2.2), Gouéré's inclusion for box paths (Lemma 3.1), the independence of the lines outside a direction window of width $O(1/M)$ between two regions at distance $M\alpha$, and a level drop for the window lines (Lemma 4.1), which give the recursion (4.2) and, with $M = \alpha^{1/2}$, the induction of Theorem 5.2.
3. The level drop costs ten levels per scale. A subcritical phase would follow from the bound (6.2) on the covariance term (6.1) at a fixed level, which requires a one-arm estimate and a disjointness statement for the two arms at a site that needs a shared line.
4. One deterministic line does not cost a level in the recursion: its effect on the bounds is contracted by a factor of order $M^{-1/2}$ per scale (Proposition 7.2). A uniform bound over several lines does not close. By the Poincaré inequality, the drift-free recursion (7.4) follows from the two-replica one-line estimate (7.5), which is open.

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Use of AI tools

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